

Vivian Huynh & Levi Hood

Red Light Therapy & Mental Health

INFO 526 - Spring 2026



College of
Information Science

OVERVIEW

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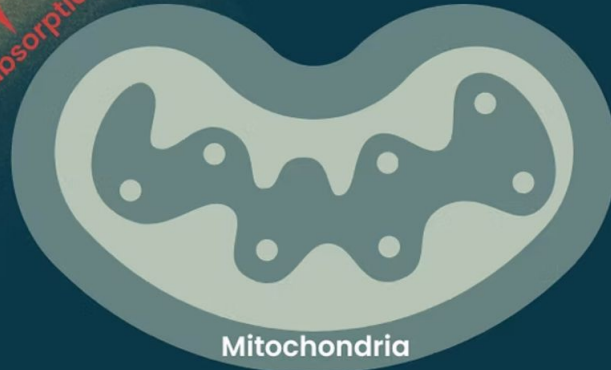
WHAT IS PHOTOBIOMODULATION THERAPY?

HOW DOES RED/NIR LIGHT THERAPY WORK?

RED/NIR LIGHT THERAPY HELPS THE MITOCHONDRIA WORK MORE EFFECTIVELY BY INCREASING THE EFFICIENCY OF AEROBIC RESPIRATION. THIS CAUSES MORE ATP PRODUCTION, WHICH FACILITATES CELLULAR PROCESSES ESSENTIAL FOR HEALING AND PROMOTING POSITIVE HEALTH.

Red light

absorption

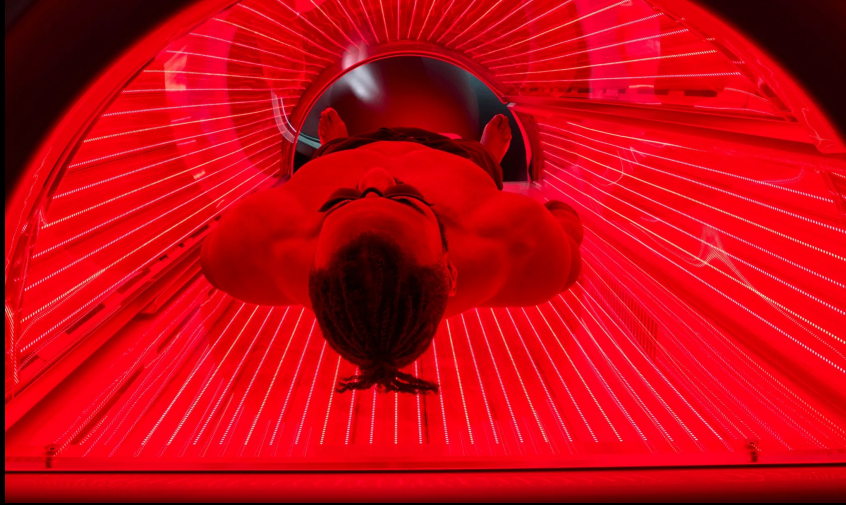


increase efficiency
production of

ATP

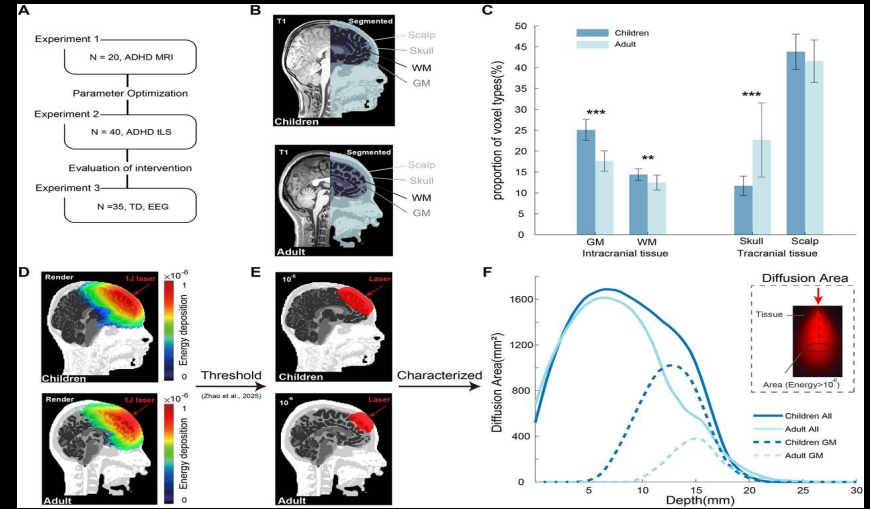


PHOTOBIOMODULATION THERAPY (PBMt)



COSMETIC = WIDESPREAD USE

- Gained significant attention for a wide range of claimed cosmetic benefits, including improved skin health and hair growth



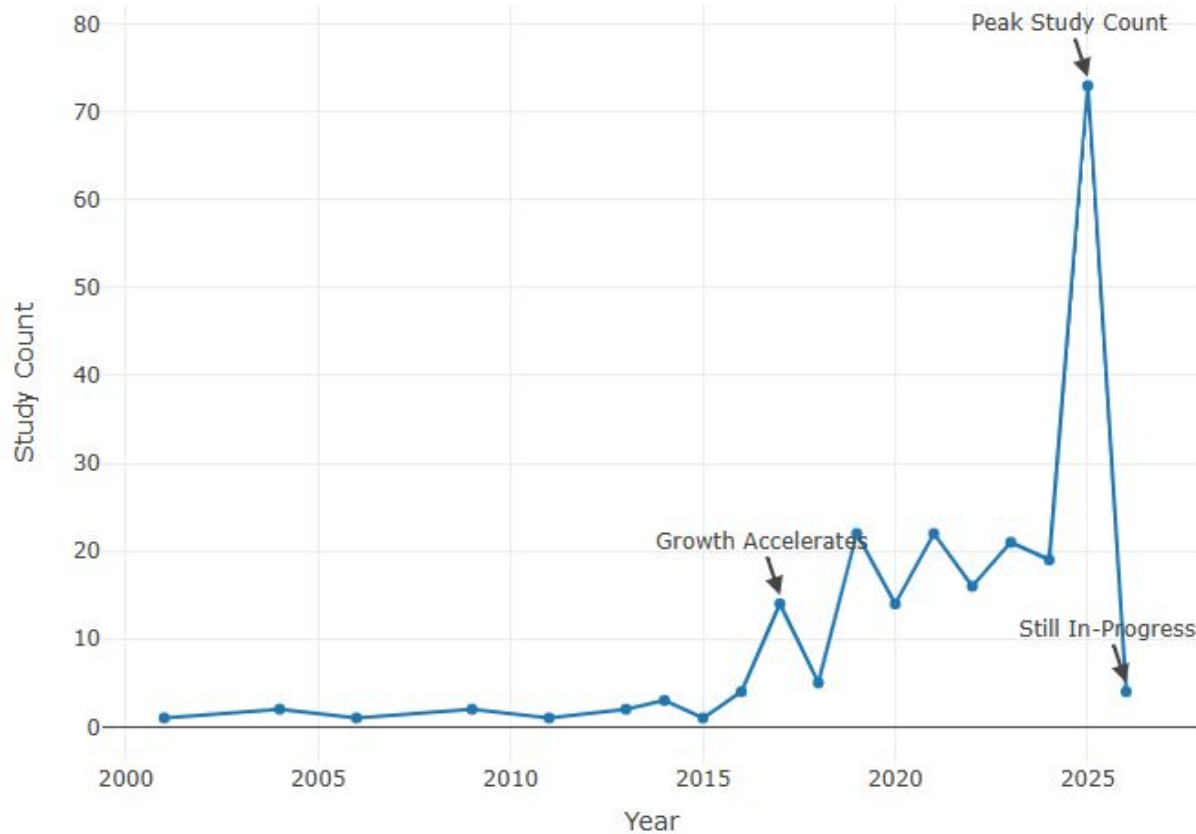
CLINICAL RESEARCH = GROWING

- Growing interest in clinical applications, such as decreased pain and inflammation, wound healing, muscle recovery, and improved cognitive function

MENTAL HEALTH APPLICATIONS

ADHD
Anxiety
Autism Symptoms
Brain Stimulation
Cognitive Decline
Dementia
Depression
Memory
Mood
Motor Skills
OCD
PTSD
Schizophrenia
Sleep Quality
Visual Memory

Growth of Mental Health Applications of PBM/RLT Research (Year-by-Year)



PROJECT GOALS

To organize and visualize the research landscape of RLT, specifically on mental health outcomes:

- Which mental health conditions have the most published research, and how do outcomes vary across them?
- Who is being studied — and does population type and age shape which conditions get research attention?
- How rigorous is the evidence base — and do stronger study designs produce different outcomes?



DATA COLLECTION

PICO Framework

- Population/Problem
- Intervention
- Comparison
- Outcome

Sources:

- PubMed, ScienceDirect, Sage Journals, etc.



DATA PREPARATION

- Using the PICO framework -> 102 study entries
- Standardized data formatting across all columns
- Expanded multi-condition and multi-outcome entries into separate rows so each study contributes independently to each condition and outcome it examined
- Consolidated inconsistent label variants using regex matching — for example merging multiple Alzheimer's spellings into one label — resulting in 26 clean condition categories and 26 outcome categories
- Final dataset: 227 rows, each representing a unique study × condition × outcome combination, expanded from the original 102 entries

```
37 #split conditions into separate rows
38 separate_longer_delim(condition, delim = ";") |>
39 mutate(condition = str_trim(condition)) |>
40 filter(condition != "", !is.na(condition)) |> #added empty string filters to catch blank rows created by splitting
41 mutate(condition = case_when(
42   str_detect(condition, regex("alzheimer|dementia.*alzheimer|alzheimer.*dementia", ignore_case = TRUE)) ~ "Alzheimer's",
43   str_detect(condition, regex("dementia", ignore_case = TRUE)) ~ "Dementia",
44   str_detect(condition, regex("mild cognitive impairment|^mci$", ignore_case = TRUE)) ~ "Mild Cognitive Impairment",
45   str_detect(condition, regex("major depress|^mdd$|^depression$", ignore_case = TRUE)) ~ "Major Depressive Disorder",
46   str_detect(condition, regex("post.stroke depression", ignore_case = TRUE)) ~ "Post-Stroke Depression",
47   str_detect(condition, regex("traumatic brain injury|brain injury|^tbi$", ignore_case = TRUE)) ~ "Brain Injury",
```

DATASET COMPOSITION

study_id	year	author	condition	population_type	population_age	study_design	application	wavelength_nm
S001	2009	Schiffer et al.	Major Depressive Disorder	clinical	adult	Open-Label Pilot	transcranial	810
S001	2009	Schiffer et al.	Major Depressive Disorder	clinical	adult	Open-Label Pilot	transcranial	810
S002	2015	Cassano et al.	Major Depressive Disorder	clinical	adult	Rct	transcranial	823
S003	2018	Cassano et al.	Major Depressive Disorder	clinical	adult	Clinical Study	transcranial	823
S004	2019	Maiello et al.	Anxiety Disorder	clinical	adult	Open-Label Pilot	transcranial	830
session_duration_min	session_count	blinding	control_type	outcome_cat	outcome_direction	sample	source	publisher
8.00	1	double-blind	placebo-controlled	Anxiety	positive	10	Psychological benefits 2 and 4 weeks after a single treatment...	PubMed
8.00	1	double-blind	placebo-controlled	Mood	positive	10	Psychological benefits 2 and 4 weeks after a single treatment...	PubMed
8.00	6	double-blind	sham-controlled	Mood	positive	4	Near-Infrared Transcranial Radiation for Major Depressive Di...	PubMed
30.00	16	double-blind	sham-controlled	Mood	positive	21	Transcranial Photobiomodulation for the Treatment of Major Depressive Disorder	PubMed
20.00	56	no blinding	open-label	Anxiety	positive	15	Transcranial Photobiomodulation with Near-Infrared Light for Major Depressive Disorder	PubMed

227 Observations : 102 Studies : 18 Features

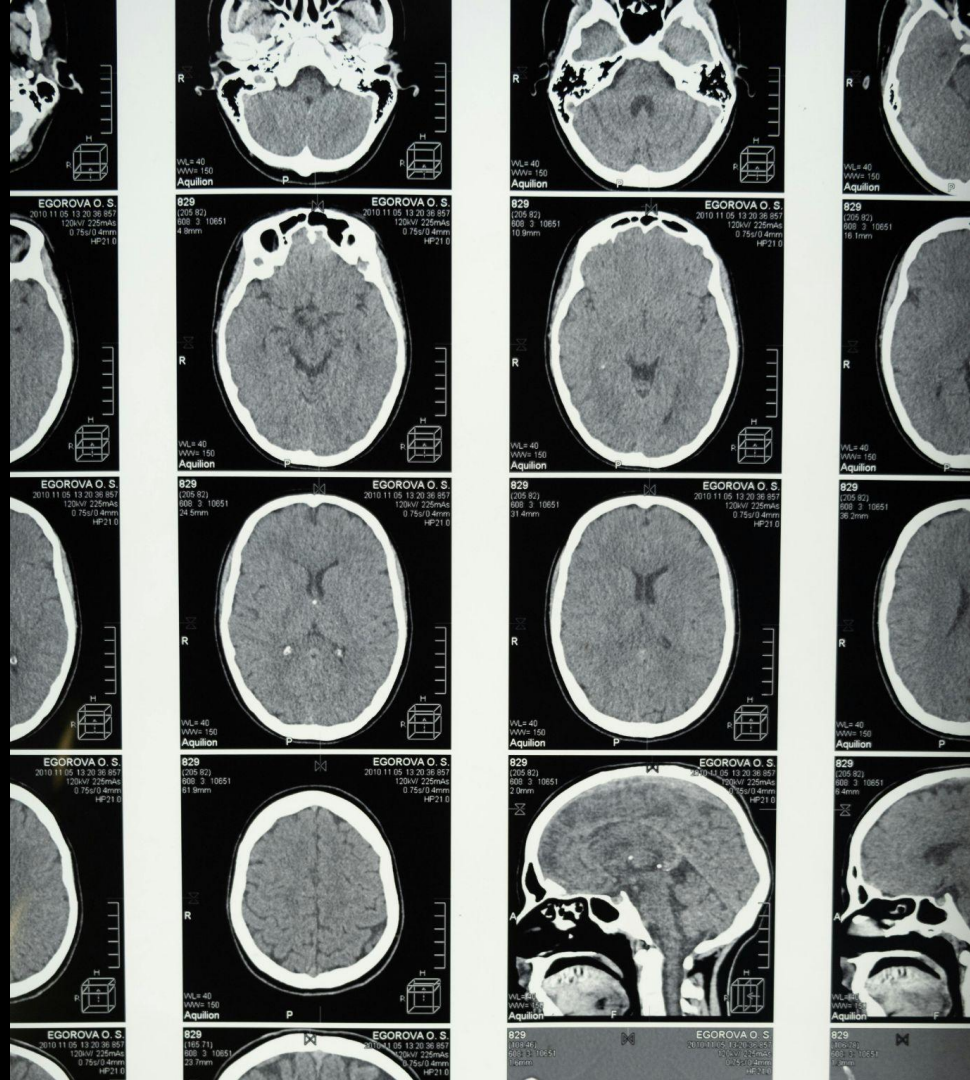
AUDIENCE & INTENT

Intent:

- To summarize and visualize the study characteristics and outcomes of RLT research from vetted sources

Audience:

- Researchers seeking to understand the scope and findings of published RLT studies
- Not necessarily the general public



DASHBOARD WALKTHROUGH

TAKEAWAYS & FUTURE CONSIDERATIONS

Selection Bias

82.4% Positive
11.8% Mixed
3.9% Inconclusive
2% Negative

Publication Count
102 = small study
count

Ways to avoid in future:

- Look for registered trials to identify completed studies that were never published
- Avoid language restrictions
- Expand dataset as research grows

Analysis Focus

- Elected to focus on an overall snapshot of what is being studied
- How: Additional page for medical specifics: wavelength, session count and min
- Why: Investors in research: background context and future projections

Technical Next Steps

- Deploy to shinyapps.io for public access
- Automate data collection using the PubMed API via rentrez R package.
- Replace manual regex cleaning with a lookup table
- Add a backend (SQL) for faster filtering as dataset grows

Additional Limitations

- No effect sizes recorded
- No funding source information
- No control for study overlap

Red Light Therapy and Mental Health

Questions?

THANK YOU,
- LEMON PEPPA STEPPERS

